University of Florida, Department of Mathematics

Algebra First-Year Exam, May 2022, Part 2

Answer 4 questions. Write your solutions clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Prove that the following polynomials are irreducible in $\mathbb{Q}[x]$, stating clearly any theorems you wish to apply.

(a) $x^4 - 75$.

(b) $x^3 - 5x^2 + x + 5$.

(c) $x^4 - 14x^3 + 2x^2 + 21x - 7$.

2. Let R be the quotient of the ring $\mathbb{Z}[i]$ of Gaussian integers by the principal ideal $(3 + 3i)$. Show that R is finite and determine the number of its elements.

3. Let R be a commutative ring with 1 and let I, J and P be ideals in R such that $I \cap J \subseteq P$.

(a) Prove that if P is a prime ideal then either $I \subseteq P$ or $J \subseteq P$.

(b) Give an example which shows that if P is not prime then the conclusion above may not hold.

4. Let R be a ring with 1 and M a (unital) left R-module.

(a) State what it means for M to satisfy the Ascending Chain Condition (i.e. for M to be Noetherian).

(b) Prove that if M satisfies the ACC then every submodule of M is finitely generated.

(c) Give an example of a finitely generated module that does not satisfy the ACC.

5. Determine the conjugacy classes of elements of order 8 in $\mathrm{GL}(4, \mathbb{Q}(\sqrt{2}))$. (Hint: Factorize $x^4 + 1$ in $\mathbb{Q}(\sqrt{2})[x]$.)

6. Let E be an extension field of the field F.

(a) What does it mean for E to be an algebraic extension of F?

(b) Show that if α and β are elements of E that are both roots of the same irreducible polynomial in $F[x]$, then the subfields $F(\alpha)$ and $F(\beta)$ of E are isomorphic.

(c) Give an example where the subfields in (b) are not equal.